during equinoxes.

But due to atmosphere, the sun's rays gets refracted (bending of light). Refraction is particularly stronger during the morning and the evening time when the sun's rays are slant.

Even though the actual sun is below the horizon, its apparent image would appear above the horizon due to refraction. This makes the days longer than nights at the equator.

61. **Statement 1: CORRECT**—The ozonosphere is also called chemosphere because, a lot of chemical activity goes on here. It lies at an altitude between 30 km and 60 km from the earth's surface and spans the stratosphere and lower mesosphere.

    Statement 2: INCORRECT—Noctilucent cloud is a luminous cloud of a kind occasionally seen at night in summer in high latitudes, at the altitude of the mesopause.

62. **Statement 1: INCORRECT**—Aurora, sometimes referred to as polar lights, northern lights (aurora borealis), or southern lights (aurora australis), is a natural light display in the Earth's sky, predominantly seen in the high-latitude regions (around the Arctic and Antarctic). Auroras are not clouds but glowing lights at mid nights.

    Statement 2: CORRECT—Auroras are the result of disturbances in the magnetosphere caused by solar wind.

    Statement 1: CORRECT—A geomagnetic storm causes the auroral ovals (north and south) to expand, bringing the aurora to lower latitudes.

63. **Statement 1: INCORRECT**—Albedo of land is much greater than albedo of oceans and water bodies. Especially snow covered areas reflect up to 70%-90% of insolation.

    Statement 2: INCORRECT—Average penetration of sunlight is more in water-up to 20 metres, than in land - where it is up to 1 metre only. Therefore, land cools or becomes hot more rapidly compared to oceans.

    Statement 3: CORRECT—The specific heat of water is 2.5 times higher than landmass, therefore water takes longer to get heated up and to cool down.

64. **Statement 1: CORRECT**—Temperature gradients are usually low over the eastern margins of continents. This is because of warm